Writing Exercises

199. Priam bought a jacket that was on sale for 40% off. The original price of the jacket was $\$150$. While the sales clerk figured the price by calculating the amount of discount and then subtracting that amount from $\$150$, Priam found the price faster by calculating 60% of $\$150$.

ⓐ Explain why Priam was correct.

ⓑ Will Priam's method work for any original price?

200. Roxy bought a scarf on sale for 50% off. The original price of the scarf was $\$32.90$. Roxy claimed that the price she paid for the scarf was the same as the amount she saved. Was Roxy correct? Explain.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
solve sales tax applications.			
solve commission applications.			
solve discount applications.			
solve mark-up applications.			

ⓑ *What does this checklist tell you about your mastery of this section? What steps will you take to improve?*

6.4 Solve Simple Interest Applications

Learning Objectives

By the end of this section, you will be able to:

- Use the simple interest formula
- Solve simple interest applications

☑ **BE PREPARED** 6.9 Before you get started, take this readiness quiz.

Solve $0.6y = 45$.

If you missed this problem, review Example 5.43.

☑ **BE PREPARED** 6.10 Solve $\frac{n}{1.45} = 4.6$.

If you missed this problem, review Example 5.44.

Use the Simple Interest Formula

Do you know that banks pay you to let them keep your money? The money you put in the bank is called the **principal**, P, and the bank pays you **interest**, I. The interest is computed as a certain percent of the principal; called the **rate of interest**, r. The rate of interest is usually expressed as a percent per year, and is calculated by using the decimal equivalent of the percent. The variable for time, t, represents the number of years the money is left in the account.

Simple Interest

If an amount of money, P, the principal, is invested for a period of t years at an annual interest rate r, the amount of interest, I, earned is

$$I = Prt$$

where

$$\begin{aligned} I &= \text{interest} \\ P &= \text{principal} \\ r &= \text{rate} \\ t &= \text{time} \end{aligned}$$

Interest earned according to this formula is called **simple interest**.

The formula we use to calculate simple interest is $I = Prt$. To use the simple interest formula we substitute in the values for variables that are given, and then solve for the unknown variable. It may be helpful to organize the information by listing all four variables and filling in the given information.

EXAMPLE 6.33

Find the simple interest earned after 3 years on $\$500$ at an interest rate of 6%.

Solution

Organize the given information in a list.

$$\begin{aligned} I &= ? \\ P &= \$500 \\ r &= 6\% \\ t &= 3 \text{ years} \end{aligned}$$

We will use the simple interest formula to find the interest.

Write the formula.	$I = Prt$
Substitute the given information. Remember to write the percent in decimal form.	$I = (500)\,(0.06)\,(3)$
Simplify.	$I = 90$
Check your answer. Is \$90 a reasonable interest earned on \$500 in 3 years?	
In 3 years the money earned 18%. If we rounded to 20%, the interest would have been 500(0.20) or \$100. Yes, \$90 is reasonable.	
Write a complete sentence that answers the question.	The simple interest is \$90.

TRY IT 6.65 Find the simple interest earned after 4 years on $\$800$ at an interest rate of 5%.

TRY IT 6.66 Find the simple interest earned after 2 years on $\$700$ at an interest rate of 4%.

In the next example, we will use the simple interest formula to find the principal.

EXAMPLE 6.34

Find the principal invested if $\$178$ interest was earned in 2 years at an interest rate of 4%.

Solution

Organize the given information in a list.

$I = \$178$
$P = ?$
$r = 4\%$
$t = 2$ years

We will use the simple interest formula to find the principal.

Write the formula.	$I = Prt$
Substitute the given information.	$178 = P(0.04)(2)$
Divide.	$\frac{178}{0.08} = \frac{0.08P}{0.08}$
Simplify.	$2{,}225 = P$
Check your answer. Is it reasonable that $2,225 would earn $178 in 2 years?	
$I = Prt$	
$178 \stackrel{?}{=} 2{,}225(0.04)(2)$	
$178 = 178$✓	
Write a complete sentence that answers the question.	The principal is $2,225.

TRY IT 6.67 Find the principal invested if $\$495$ interest was earned in 3 years at an interest rate of 6%.

TRY IT 6.68 Find the principal invested if $\$1{,}246$ interest was earned in 5 years at an interest rate of 7%.

Now we will solve for the rate of interest.

EXAMPLE 6.35

Find the rate if a principal of $\$8{,}200$ earned $\$3{,}772$ interest in 4 years.

Solution

Organize the given information.

$I = \$3{,}772$
$P = \$8{,}200$
$r = ?$
$t = 4$ years

We will use the simple interest formula to find the rate.

Write the formula.	$I = Prt$
Substitute the given information.	$3{,}772 = 8{,}200r(4)$
Multiply.	$3{,}772 = 32{,}800r$

Divide.	$\frac{3,772}{32,800} = \frac{32,800r}{32,800}$
Simplify.	$0.115 = r$
Write as a percent.	$11.5\% = r$
Check your answer. Is 11.5% a reasonable rate if \$3,772 was earned in 4 years?	
$I = Prt$	
$3,772 \stackrel{?}{=} 8,200\,(0.115)\,(4)$	
$3,772 = 3,772$✓	
Write a complete sentence that answers the question.	The rate was 11.5%.

TRY IT 6.69 Find the rate if a principal of \$5,000 earned \$1,350 interest in 6 years.

TRY IT 6.70 Find the rate if a principal of \$9,000 earned \$1,755 interest in 3 years.

Solve Simple Interest Applications

Applications with simple interest usually involve either investing money or borrowing money. To solve these applications, we continue to use the same strategy for applications that we have used earlier in this chapter. The only difference is that in place of translating to get an equation, we can use the simple interest formula.

We will start by solving a simple interest application to find the interest.

EXAMPLE 6.36

Nathaly deposited \$12,500 in her bank account where it will earn 4% interest. How much interest will Nathaly earn in 5 years?

Solution

We are asked to find the Interest, I.

Organize the given information in a list.

$I = ?$
$P = \$12,500$
$r = 4\%$
$t = 5$ years

Write the formula.	$I = Prt$
Substitute the given information.	$I = (12,500)\,(0.04)\,(5)$
Simplify.	$I = 2,500$
Check your answer. Is \$2,500 a reasonable interest on \$12,500 over 5 years?	

At 4% interest per year, in 5 years the interest would be 20% of the principal. Is 20% of $12,500 equal to $2,500? Yes.	
Write a complete sentence that answers the question.	The interest is $2,500.

TRY IT 6.71 Areli invested a principal of $950 in her bank account with interest rate 3%. How much interest did she earn in 5 years?

TRY IT 6.72 Susana invested a principal of $36,000 in her bank account with interest rate 6.5%. How much interest did she earn in 3 years?

There may be times when you know the amount of interest earned on a given principal over a certain length of time, but you don't know the rate. For instance, this might happen when family members lend or borrow money among themselves instead of dealing with a bank. In the next example, we'll show how to solve for the rate.

EXAMPLE 6.37

Loren lent his brother $3,000 to help him buy a car. In 4 years his brother paid him back the $3,000 plus $660 in interest. What was the rate of interest?

Solution

We are asked to find the rate of interest, r.

Organize the given information.

$$\begin{aligned} I &= 660 \\ P &= \$3{,}000 \\ r &= ? \\ t &= 4 \text{ years} \end{aligned}$$

Write the formula.	$I = Prt$
Substitute the given information.	$660 = (3{,}000)\, r(4)$
Multiply.	$660 = (12{,}000)\, r$
Divide.	$\frac{660}{12{,}000} = \frac{(12{,}000)r}{12{,}000}$
Simplify.	$0.055 = r$
Change to percent form.	$5.5\% = r$
Check your answer. Is 5.5% a reasonable interest rate to pay your brother?	
$I = Prt$	
$660 \stackrel{?}{=} (3{,}000)\,(0.055)\,(4)$	

$660 = 660$✓	
Write a complete sentence that answers the question.	The rate of interest was 5.5%.

TRY IT 6.73 Jim lent his sister $\$5{,}000$ to help her buy a house. In 3 years, she paid him the $\$5{,}000$, plus $\$900$ interest. What was the rate of interest?

TRY IT 6.74 Hang borrowed $\$7{,}500$ from her parents to pay her tuition. In 5 years, she paid them $\$1{,}500$ interest in addition to the $\$7{,}500$ she borrowed. What was the rate of interest?

There may be times when you take a loan for a large purchase and the amount of the principal is not clear. This might happen, for instance, in making a car purchase when the dealer adds the cost of a warranty to the price of the car. In the next example, we will solve a simple interest application for the principal.

EXAMPLE 6.38

Eduardo noticed that his new car loan papers stated that with an interest rate of 7.5%, he would pay $\$6{,}596.25$ in interest over 5 years. How much did he borrow to pay for his car?

Solution

We are asked to find the principal, P.

Organize the given information.

$I = 6{,}596.25$
$P = ?$
$r = 7.5\%$
$t = 5$ years

Write the formula.	$I = Prt$
Substitute the given information.	$6{,}596.25 = P(0.075)(5)$
Multiply.	$6{,}596.25 = 0.375P$
Divide.	$\frac{6{,}596.25}{0.375} = \frac{0.375P}{0.375}$
Simplify.	$17{,}590 = P$
Check your answer. Is $17,590 a reasonable amount to borrow to buy a car?	
$I = Prt$	
$6{,}596.25 \stackrel{?}{=} (17{,}590)(0.075)(5)$	
$6{,}596.25 = 6{,}596.25$✓	
Write a complete sentence that answers the question.	The amount borrowed was $17,590.

TRY IT 6.75 Sean's new car loan statement said he would pay $\$4{,}866.25$ in interest from an interest rate of 8.5% over 5 years. How much did he borrow to buy his new car?

TRY IT 6.76 In 5 years, Gloria's bank account earned $\$2{,}400$ interest at 5%. How much had she deposited in the account?

In the simple interest formula, the rate of interest is given as an annual rate, the rate for one year. So the units of time must be in years. If the time is given in months, we convert it to years.

EXAMPLE 6.39

Caroline got $\$900$ as graduation gifts and invested it in a 10-month certificate of deposit that earned 2.1% interest. How much interest did this investment earn?

Solution

We are asked to find the interest, I.

Organize the given information.

$$\begin{aligned} I &= ? \\ P &= \$900 \\ r &= 2.1\% \\ t &= 10 \text{ months} \end{aligned}$$

Write the formula.	$I = Prt$
Substitute the given information, converting 10 months to $\frac{10}{12}$ of a year.	$I = \$900(0.021)\left(\frac{10}{12}\right)$
Multiply.	$I = 15.75$
Check your answer. Is $15.75 a reasonable amount of interest?	
If Caroline had invested the $900 for a full year at 2% interest, the amount of interest would have been $18. Yes, $15.75 is reasonable.	
Write a complete sentence that answers the question.	The interest earned was $15.75.

TRY IT 6.77 Adriana invested $\$4{,}500$ for 8 months in an account that paid 1.9% interest. How much interest did she earn?

TRY IT 6.78 Milton invested $\$2{,}460$ for 20 months in an account that paid 3.5% interest How much interest did he earn?

SECTION 6.4 EXERCISES

Practice Makes Perfect

Use the Simple Interest Formula

In the following exercises, use the simple interest formula to fill in the missing information.

201.

Interest	Principal	Rate	Time (years)
	$1200	3%	5

Table 6.5

202.

Interest	Principal	Rate	Time (years)
	$1500	2%	4

Table 6.6

203.

Interest	Principal	Rate	Time (years)
$4410		4.5%	7

Table 6.7

204.

Interest	Principal	Rate	Time (years)
$2112		3.2%	6

Table 6.8

205.

Interest	Principal	Rate	Time (years)
$577.08	$4580		2

Table 6.9

206.

Interest	Principal	Rate	Time (years)
$528.12	$3260		3

Table 6.10

In the following exercises, solve the problem using the simple interest formula.

207. Find the simple interest earned after 5 years on $600 at an interest rate of 3%.

208. Find the simple interest earned after 4 years on $900 at an interest rate of 6%.

209. Find the simple interest earned after 2 years on $8,950 at an interest rate of 3.24%.

210. Find the simple interest earned after 3 years on $6,510 at an interest rate of 2.85%.

211. Find the simple interest earned after 8 years on $15,500 at an interest rate of 11.425%.

212. Find the simple interest earned after 6 years on $23,900 at an interest rate of 12.175%.

213. Find the principal invested if $656 interest was earned in 5 years at an interest rate of 4%.

214. Find the principal invested if $177 interest was earned in 2 years at an interest rate of 3%.

215. Find the principal invested if $70.95 interest was earned in 3 years at an interest rate of 2.75%.

216. Find the principal invested if $636.84 interest was earned in 6 years at an interest rate of 4.35%.

217. Find the principal invested if $15,222.57 interest was earned in 6 years at an interest rate of 10.28%.

218. Find the principal invested if $10,953.70 interest was earned in 5 years at an interest rate of 11.04%.

219. Find the rate if a principal of $5,400 earned $432 interest in 2 years.

220. Find the rate if a principal of $2,600 earned $468 interest in 6 years.

221. Find the rate if a principal of $11,000 earned $1,815 interest in 3 years.

222. Find the rate if a principal of $8,500 earned $3,230 interest in 4 years.

Solve Simple Interest Applications

In the following exercises, solve the problem using the simple interest formula.

223. Casey deposited $1,450 in a bank account with interest rate 4%. How much interest was earned in 2 years?

224. Terrence deposited $5,720 in a bank account with interest rate 6%. How much interest was earned in 4 years?

225. Robin deposited $31,000 in a bank account with interest rate 5.2%. How much interest was earned in 3 years?

226. Carleen deposited $16,400 in a bank account with interest rate 3.9%. How much interest was earned in 8 years?

227. Hilaria borrowed $8,000 from her grandfather to pay for college. Five years later, she paid him back the $8,000, plus $1,200 interest. What was the rate of interest?

228. Kenneth lent his niece $1,200 to buy a computer. Two years later, she paid him back the $1,200, plus $96 interest. What was the rate of interest?

229. Lebron lent his daughter $20,000 to help her buy a condominium. When she sold the condominium four years later, she paid him the $20,000, plus $3,000 interest. What was the rate of interest?

230. Pablo borrowed $50,000 to start a business. Three years later, he repaid the $50,000, plus $9,375 interest. What was the rate of interest?

231. In 10 years, a bank account that paid 5.25% earned $18,375 interest. What was the principal of the account?

232. In 25 years, a bond that paid 4.75% earned $2,375 interest. What was the principal of the bond?

233. Joshua's computer loan statement said he would pay $1,244.34 in interest for a 3 year loan at 12.4%. How much did Joshua borrow to buy the computer?

234. Margaret's car loan statement said she would pay $7,683.20 in interest for a 5 year loan at 9.8%. How much did Margaret borrow to buy the car?

235. Caitlin invested $8,200 in an 18-month certificate of deposit paying 2.7% interest. How much interest did she earn form this investment?

236. Diego invested $6,100 in a 9-month certificate of deposit paying 1.8% interest. How much interest did he earn form this investment?

237. Airin borrowed $3,900 from her parents for the down payment on a car and promised to pay them back in 15 months at a 4% rate of interest. How much interest did she owe her parents?

238. Yuta borrowed $840 from his brother to pay for his textbooks and promised to pay him back in 5 months at a 6% rate of interest. How much interest did Yuta owe his brother?

Everyday Math

239. **Interest on savings** Find the interest rate your local bank pays on savings accounts.
ⓐ What is the interest rate?
ⓑ Calculate the amount of interest you would earn on a principal of $8,000 for 5 years.

240. **Interest on a loan** Find the interest rate your local bank charges for a car loan.
ⓐ What is the interest rate?
ⓑ Calculate the amount of interest you would pay on a loan of $8,000 for 5 years.